Oxyile Protocol - Detailed FinTech Enterprise Architecture

FLOW 1: Upload Docs

live verification updates

live status updates

FLOW 1: Live Alert

Supabase Ecosystem - PostgreSQL

Enforces Row Level Security (RLS) and broadcasts Realtime WebSockets to update screens.

Storage Buckets (RC & ID Docs)

Realtime WebSockets

postgres_changes

Auth + RLS Policies

persist state

Core DB (Users, Handshakes)

upload docs

persist tx hash

FLOW 3: Tx Hash

External APIs & Blockchains

Fiat movement, immutable ledgers, and emails.

insert prefs

FLOW 4: Store Data

Polygon Amoy Testnet (Immutable Ledger)

update LTV

broadcast tx

signed URL

FLOW 3: Hash + Mint

Resend API (Emails & Alerts)

confirmed tx hash

Frontend UI Layer - Next.js (Vercel)

Captures Waitlist data, secure document uploads, and JIT funding actions.

FLOW 2: Funds in Escrow

completion callback

Borrower Dashboard (Collateral, Mandates)

FLOW 2: GoCardless JIT

GoCardless API (Sandbox Escrow)

FLOW 3: Webhook

Platform Actors

Investors fund escrow, Borrowers provide collateral, and Admins oversee compliance.

actions

Backend Core - API & Server Actions

Handles fiat math, routes JIT funds, enforces LTV, and signs Web3 txs.

checkout URL

Risk Management & Web3

Identity & Escrow Flow

create mandate

funds escrow

uploads docs

Investor

Investor Marketplace (JIT Escrow)

trigger relayer

anchor call

Relayer Service (Web3 Keys)

Transaction Signer

checkout

State Machine (PENDING->FUNDED)

verifies collateral

Platform Admin

Admin Portal (Realtime verifications)

completeBorrowerBankLink()

FLOW 2: Payout Routing

Waitlist & Compliance

verifyAsset() (70% LTV)

export

Borrower

API Edge Router

submits answers

Success Callbacks (/handshake/success)

callbacks

form submit

FLOW 2: "Fund Escrow"

confirmEscrowAndRoute()

PDF Generator

FLOW 4: Compliance Capture

Waitlist UI (Open Banking)

initiateJITFunding()

Platform Reliability

Event Stream

Monitoring

FLOW 1: Verify Asset

FLOW 4: PDF Export